

Distributive Fairness through Randomized Scheduling in JADE

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Abstract

We study fairness properties of randomized instruction-to-thread scheduling for a low-level (hardware-level) instruction set (for example, assembly-level instructions inside a binary) executed in parallel on n threads. We define *absolute fairness* and *distributive fairness*, show that absolute fairness implies distributive fairness, and then study limits and sufficient conditions for obtaining distributive fairness under randomized scheduling. The main technical tool is McDiarmid’s bounded differences inequality applied to the sample mean of thread works. Please note that these results are purely theoretical, to show the limitations of randomized scheduling. These limitations are not specific to JADE, but to any method of randomized scheduling that distributes instructions over threads in a randomized automated manner. Because of these limitations, we, at JADE, have come up with other approximations for distributed scheduling that are an upgrade over pure randomized scheduling. This paper, written by us, serves as the mathematical foundation for such optimizations.

1 Informal statement of the main idea

Suppose a program consists of many small instructions; each instruction contributes some bounded amount of *work* (measured in uniform units). We wish to assign these instructions (randomly) to n worker threads $T_1, \dots, T_n$. Let X_i denote the total work assigned to thread T_i . The objective is to make the distribution of $\{X_i\}$ *fair* in some sense — we will formalize two such notions. The paper studies how (and when) randomized scheduling yields fairness.

2 Mathematical setup and notation

- Let the *total program work* be p units. We assume the unit is chosen so $p > 0$.
- There are n threads $T_1, \dots, T_n$. Their observed (random) works are $X_1, X_2, \dots, X_n$ satisfying

$$\sum_{i=1}^n X_i = p.$$

- We treat $X_1, \dots, X_{n-1}$ as i.i.d. random variables sampled by the randomized scheduling procedure from an unknown distribution with (finite) mean μ . We do *not* assume independence of X_n with the others; in fact $X_n = p - \sum_{i=1}^{n-1} X_i$ so it is determined by them.
- We assume all thread works are bounded in $[0, p]$. (This is natural when p is the total work.)

3 Fairness definitions

Definition 3.1 (Absolute fairness). *Fix $\varepsilon > 0$ and $\delta \in (0, 1)$. We say the randomized scheduler provides (ε, δ) absolute fairness if for every $i = 1, \dots, n-1$,*

$$\Pr(|X_i - \mu| \geq \varepsilon) \leq \delta.$$

Intuitively: each individual thread's work is within ε of μ , which is the mean of the distribution from which these threads are assumed to belong, with probability at least $1 - \delta$.

Definition 3.2 (Distributive fairness). *Fix $\varepsilon > 0$ and $\delta \in (0, 1)$. Let*

$$\bar{X} = \frac{1}{n-1} \sum_{i=1}^{n-1} X_i$$

be the sample mean of the first $n-1$ (i.i.d.) thread works. We say the randomized scheduler provides (ε, δ) distributive fairness if

$$\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq \delta.$$

Intuitively: the sample mean of the $n-1$ threads is within ε of the true mean with probability at least $1 - \delta$.

4 Relationship between the two notions

Lemma 4.1 (Absolute fairness implies distributive fairness (via union bound)). *Assume $X_1, \dots, X_{n-1}$ satisfy $\Pr(|X_i - \mu| \geq \varepsilon) \leq \delta$ for each $i = 1, \dots, n-1$. Then*

$$\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq (n-1)\delta.$$

In particular, (ε, δ') -distributive fairness follows with $\delta' = (n-1)\delta$.

Proof. If for every i we have $|X_i - \mu| < \varepsilon$, then by averaging,

$$|\bar{X} - \mu| = \left| \frac{1}{n-1} \sum_{i=1}^{n-1} (X_i - \mu) \right| \leq \frac{1}{n-1} \sum_{i=1}^{n-1} |X_i - \mu| < \varepsilon.$$

Hence the event $\{|\bar{X} - \mu| \geq \varepsilon\}$ is contained in the union $\bigcup_{i=1}^{n-1} \{|X_i - \mu| \geq \varepsilon\}$. By the union bound,

$$\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq \sum_{i=1}^{n-1} \Pr(|X_i - \mu| \geq \varepsilon) \leq (n-1)\delta.$$

This completes the proof. □

Remark 4.2. *The lemma shows that absolute fairness with per-thread failure probability δ implies distributive fairness with failure probability at most $(n-1)\delta$. If one wants the same δ for distributive fairness, it is sufficient to guarantee per-thread absolute fairness with $\delta/(n-1)$.*

5 Concentration of the sample mean: McDiarmid's inequality

We will use McDiarmid's bounded difference inequality (a standard tool for concentration of functions of independent variables).

Theorem 5.1 (McDiarmid, bounded differences). *Let $Y_1, \dots, Y_m$ be independent random variables taking values in sets $\mathcal{Y}_1, \dots, \mathcal{Y}_m$. Let $f : \mathcal{Y}_1 \times \dots \times \mathcal{Y}_m \rightarrow \mathbb{R}$ satisfy the bounded difference property: for each i there exists c_i such that for all arguments and for any alternate value y'_i ,*

$$|f(y_1, \dots, y_i, \dots, y_m) - f(y_1, \dots, y'_i, \dots, y_m)| \leq c_i.$$

Then for every $\varepsilon > 0$,

$$\Pr(|f(Y) - \mathbb{E}f(Y)| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^m c_i^2}\right).$$

We will apply this theorem to the sample mean

$$Z(X_1, \dots, X_{n-1}) = \bar{X} = \frac{1}{n-1} \sum_{i=1}^{n-1} X_i,$$

which is a function of $m = n - 1$ independent random variables.

6 Theorem 1 (negative result): randomized scheduling may fail to give distributive fairness

Theorem 6.1. *For randomized assignment of instructions to threads (with only the boundedness assumption $X_i \in [0, p]$), there is no nontrivial guarantee that distributive fairness (ε, δ) holds unless the number of threads n grows as*

$$n - 1 \geq \frac{p^2}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right).$$

In other words, to force the sample mean $\bar{X}$ within ε of μ with probability at least $1 - \delta$ using only the boundedness $[0, p]$, one needs $n = \Omega(p^2/\varepsilon^2 \cdot \log(1/\delta))$, which is generally impractical because n cannot reasonably exceed p in the natural discrete-instruction interpretation.

Proof. Apply McDiarmid to $Z = \bar{X}$ with $m = n - 1$ independent inputs $X_1, \dots, X_{n-1}$.

Compute the bounded differences c_i . If we replace X_i by an independent copy X'_i (same marginal distribution), all other coordinates remain equal. The change in Z is

$$|Z(X_1, \dots, X_i, \dots, X_{n-1}) - Z(X_1, \dots, X'_i, \dots, X_{n-1})| = \frac{1}{n-1} |X_i - X'_i|.$$

Since $X_i, X'_i \in [0, p]$, we have $|X_i - X'_i| \leq p$. Therefore we can take

$$c_i = \frac{p}{n-1} \quad \text{for each } i.$$

Compute $\sum c_i^2$. There are $m = n - 1$ such terms, so

$$\sum_{i=1}^{n-1} c_i^2 = (n-1) \cdot \left(\frac{p}{n-1}\right)^2 = \frac{p^2}{n-1}.$$

Apply McDiarmid. For any $\varepsilon > 0$,

$$\Pr(|\bar{X} - \mathbb{E}\bar{X}| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum c_i^2}\right) = 2 \exp\left(-\frac{2\varepsilon^2}{p^2/(n-1)}\right) = 2 \exp\left(-\frac{2\varepsilon^2(n-1)}{p^2}\right).$$

Note $\mathbb{E}\bar{X} = \mu$ by linearity of expectation.

Solve for n to obtain a target failure probability δ . We require

$$2 \exp\left(-\frac{2\varepsilon^2(n-1)}{p^2}\right) \leq \delta.$$

Divide both sides by 2 and take natural logs (noting $\delta/2 \in (0, 1)$):

$$-\frac{2\varepsilon^2(n-1)}{p^2} \leq \ln\left(\frac{\delta}{2}\right) = -\ln\left(\frac{2}{\delta}\right).$$

Multiply by -1 and rearrange:

$$\frac{2\varepsilon^2(n-1)}{p^2} \geq \ln\left(\frac{2}{\delta}\right),$$

hence

$$n-1 \geq \frac{p^2}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right).$$

This completes the derivation.

Interpretation and conclusion. When the only assumption is the trivial bound $X_i \in [0, p]$, the number of threads required to obtain a given (ε, δ) distributive fairness scales as $\Omega(p^2/\varepsilon^2 \cdot \log(1/\delta))$. If p corresponds to the total number of indivisible instruction-units, requiring $n > p$ is meaningless (some threads would have zero instructions on average), therefore randomized scheduling without additional structural assumptions cannot in general guarantee distributive fairness. $\square$

7 Theorem 2 (positive result under stronger regularity)

Theorem 7.1. *Suppose there exists a sequence c_n with $c_n = o(1)$ as $n \rightarrow \infty$, and assume the randomized scheduler and program are such that for every i ,*

$$|X_i - X'_i| \leq c_n \frac{p}{\sqrt{n}}$$

almost surely, where X'_i is an independent copy of X_i . Then the sample mean $\bar{X}$ concentrates and to ensure $\Pr(|\bar{X} - \mu| \geq \varepsilon) \leq \delta$ it suffices to take

$$n \gtrsim \frac{c_n^2 p^2}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right),$$

and because $c_n = o(1)$ the needed n can be asymptotically smaller in terms of p than the bound in Theorem 1; in particular one can achieve distributive fairness without forcing n to exceed the natural scale p .

Proof. We again apply McDiarmid to $Z = \bar{X}$. Under the stated stronger regularity we have the replacement bound

$$|X_i - X'_i| \leq c_n \frac{p}{\sqrt{n}}.$$

Thus the bounded difference constant for coordinate i can be taken as

$$c_i = \frac{1}{n-1} |X_i - X'_i| \leq \frac{1}{n-1} c_n \frac{p}{\sqrt{n}} = c_n \frac{p}{(n-1)\sqrt{n}}.$$

Now compute the sum of squares:

$$\sum_{i=1}^{n-1} c_i^2 \leq (n-1) \cdot \left(c_n \frac{p}{(n-1)\sqrt{n}} \right)^2 = (n-1) \cdot c_n^2 \frac{p^2}{(n-1)^2 n} = c_n^2 \frac{p^2}{(n-1)n}.$$

For large n we may approximate $(n-1)n \approx n^2$, giving the reasonable asymptotic bound

$$\sum_{i=1}^{n-1} c_i^2 \lesssim c_n^2 \frac{p^2}{n^2}.$$

Apply McDiarmid:

$$\Pr(|\overline{X} - \mu| \geq \varepsilon) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum c_i^2}\right) \lesssim 2 \exp\left(-\frac{2\varepsilon^2 n^2}{c_n^2 p^2}\right).$$

Set the right-hand side $\leq \delta$, take logs and rearrange:

$$-\frac{2\varepsilon^2 n^2}{c_n^2 p^2} \leq \ln\left(\frac{\delta}{2}\right) \implies \frac{2\varepsilon^2 n^2}{c_n^2 p^2} \geq \ln\left(\frac{2}{\delta}\right).$$

Hence

$$n^2 \gtrsim \frac{c_n^2 p^2}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right),$$

and therefore

$$n \gtrsim \frac{c_n p}{\sqrt{2}\varepsilon} \sqrt{\ln\left(\frac{2}{\delta}\right)}.$$

Since $c_n = o(1)$, the multiplicative factor $c_n p$ can be asymptotically much smaller than p itself for large n (or for programs with structure that enforces small c_n). Thus under this stronger regularity condition the thread-count required for distributive fairness can be significantly smaller than the pessimistic $\Theta(p^2)$ scaling from Theorem 1, and it is possible to obtain distributive fairness without allocating frivolously many threads. $\square$

Remark 7.2. *The condition $|X_i - X'_i| \leq c_n p / \sqrt{n}$ is a way to formalize that the per-thread fluctuation (between two independent samplings) is sublinear in p and shrinks with n . It can arise when each thread receives many small independent instruction-units so that per-thread totals concentrate by themselves (a law-of-large-numbers style effect), or when the randomized scheduler's assignment mechanism controls variance tightly.*

8 Discussion and implications for JADE's randomized scheduler

- *Negative message (Theorem 1):* Without additional regularity on how instructions map to work contributions, randomized scheduling alone cannot guarantee distributive fairness unless one is willing to provision an impractically large number of threads (scaling like p^2 for fixed ε, δ). In practical systems this is infeasible.
- *Positive message (Theorem 2):* If the scheduling mechanism or program structure ensures that per-thread totals do not vary too widely (quantified by a small c_n factor), then randomized scheduling yields good distributive fairness with far fewer threads: the number of threads needed grows only linearly with p times a small factor c_n (and with $1/\varepsilon$ and $\sqrt{\log(1/\delta)}$).
- *Practical implication for JADE:* To get fairness via randomization a practical path is to enforce or encourage the regularity assumption: (i) break instructions into many small uniform units, (ii) use randomized assignment schemes that reduce per-thread variance (e.g., round-robin with random offsets, power-of-two-choices variants, or hashing combined with rebalancing), or (iii) employ light-weight local balancing that keeps $|X_i - X_j|$ small relative to $p/\sqrt{n}$.

9 Conclusion

We provided a formal statement of two fairness notions, proved that absolute fairness implies distributive fairness (with a quantitative union-bound), and used McDiarmid’s inequality to (i) show an impossibility-type scaling when only boundedness is assumed, and (ii) show sufficiency under stronger per-thread regularity conditions. For JADE’s optimizer, these results suggest focusing efforts on variance-reducing scheduling primitives or finer instruction decomposition to achieve distributive fairness in practice.